

TRIGONOMETRIC INTEGRATION: ADVANCED PROBLEM SHEET

Complete the problems below. Solutions and hints are available in a separate pdf, with marking available upon request.

Note: these problems focus only on the raw mechanics of integrating. Exam-style problems typically involve more problem solving.

1. Consider the following integral

$$\int 2 \sin(x) \cos(x) dx.$$

- (a) Solve the integral by using the double angle formula.
- (b) Solve the integral by making the substitution $u = \sin(x)$.
- (c) Explain how these answers are equivalent.

2. Solve the following integrals.

- (a) $\int \sin^2(x) dx$,
- (b) $\int \cos^2(x) dx$,
- (c) $\int \tan^2(x) dx$.

3. Solve the following integrals.

- (a) $\int \operatorname{cosec}^7(x) \cot(x) dx$,
- (b) $\int x \cos(3x) dx$.

4. Solve the following integrals.

- (a) $\int e^x \sin(x) dx$,
- (b) $\int \tan(x) dx$,
- (c) $\int \tan^6(x) \sec^2(x) dx$.

5. Solve the following integrals.

- (a) $\int \sin^3(x) dx$,
- (b) $\int x \sec^2(x) dx$,
- (c) $\int \sin^2(x) \cos^2(x) dx$.